

Jordyn O'Brien

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Education

Memorial University of Newfoundland, BEng

Sept 2021 – May 2026

- Computer Engineering Co-op
- GPA: 3.95/4.00
- Relevant Courses: Computer Security, Communication Networks, Communication Principles, Probability & Random Processes, Digital Signal Processing

Research Interests

Anonymous communications, Privacy-preserving technologies, Communication networks, Network security

Experience

Software Engineering Intern

Sep 2023 - Dec 2023, Sep 2025 - Present

Avalon Holographics

St. John's, NL

- Added new features and fixed over 15 reported bugs in a 3D battle space simulation application built on a custom game engine, enhancing functionality and improving user experience
- Designed and implemented a user responsive map scale and compass for a 3D globe based application by monitoring the camera's height and viewing direction relative to the earth's surface using C++
- Implemented brightness, contrast, and exposure controls for Avalon's holographic displays by modifying RGB values during ray tracing according to user input using HLSL, providing a customizable viewing experience

Full Stack Developer

Jan 2025 - Apr 2025

FliteX

Toronto, ON

- Led a team of developers in building four web applications for use in the airline industry by facilitating stand-ups, sprint planning, and customer meetings
- Supervised and mentored two software development interns, providing guidance to strengthen their React and Angular skills
- Built a custom API in JavaScript using Node.js and node-postgres, enabling automated decoding of planned flight routes and generation of optimized coordinate paths
- Designed and implemented front-end features in Angular and React, integrating Leaflet and Mapbox for geospatial visualization based on customer requirements

Research/Software Developer Student

May 2024 - Aug 2024

Communications Research Centre

Ottawa, ON

- Developed a map-based web application using React with Leaflet, allowing researchers to visualize 5G network layouts
- Developed an algorithm using JavaScript to project 3D antenna pattern plots onto a 2D map allowing researchers to visualize simulation results within a map-based application
- Implemented spatial decimation and alpha shape algorithms, reducing the number of data points needed for rendering 3D plots, making plots render 4 times faster

Network Specialist Student

Jan 2023 - Apr 2023

Bell

St. John's, NL

- Automated SQL database entry with a VBA solution, eliminating manual input and saving employees 4+ hours per week
- Developed a script to sort and organize rural customer addresses, streamlining the expansion of fiber-optic internet into under-served rural communities
- Upgraded a custom Microsoft Access application for cross-compatibility with 32-bit and 64-bit systems, reducing workflow interruptions for 50+ employees while new laptops were being distributed

Extracurricular

Software Team Lead, Iceberg ASV (Student Design Team) Jun 2023 - Present

- Leading the development of autonomous navigation algorithms using ROS2 and C++ , enabling a student-built surface vehicle to complete autonomous challenges involving obstacle avoidance, docking, and maneuvering
- Attended the 2024 Njord Autonomous Ship Challenge in Norway and the 2025 RoboBoat Competition in Florida alongside other team members where we placed 5th and 11th overall

Grad Committee Representative, MUN Engineering Class of 2026 Sep 2023 - Present

- Leading the planning and execution of fundraising initiatives for the engineering class
- Coordinating graduation events, including the Iron Ring Ceremony, gala, and other class celebrations

Volunteer, Girls Who Game (Dell Technologies) Sep 2023 - Present

- Helping young girls across North America engage with STEM through gaming initiatives

Projects

Indoor Asset Tracking System for Hospital Environments Apr 2025 - Present

- Developing an application to securely track healthcare assets using BLE tags, improving efficiency by allowing nurses and staff to immediately locate equipment and each other

Analyzing the Use of Concurrency in Dictionary Attacks Jul 2025 - Aug 2025

- Implemented a dictionary-based password cracking algorithm using sequential and multi-threaded approaches to evaluate how hash algorithms (MD5, SHA-1, SHA-256, SHA-512) hold up against multi-threaded attacks
- Measured execution time and calculated speed-up and efficiency of attacks using different thread counts

TripTailor Sep 2024 - Dec 2024

- Designed and developed a travel planning platform, enabling users to create, share, and search travel itineraries across cities worldwide
- Built a microservice-based architecture using Golang and React.js, enabling modularity, fault tolerance, and efficient API interactions
- Integrated profile customization, tagging, and user interaction services to enhance recommendation accuracy and user engagement

Honors and Awards

Most Valuable Tutor Award 2025

Dean's List 2021-2025

TechNL Making Waves Innovator Scholarship 2024

Verafin Computer Engineering Scholarship 2022

Freedom Foods Employee Scholarship 2021

Teaching and Tutoring Experience

Tutor, Engineering One Success Centre Sep 2022 - Present

- Tutoring first year engineering courses including programming, math, physics, and circuit analysis
- Hosted online programming tutorials for students attending MUN's remote Grenfell Campus

Private Programming Tutor, Self Employed May 2023 - Present

- Providing one-on-one tutoring in introductory programming courses
- Maintaining a 100% pass rate